



21. (a) When vapour pressure of solvent decreases, then the boiling point of solvent increases.

22. (a) **Temperature but not on volume**

**Explanation:**

Vapour pressure depends only on temperature. As temperature increases, more molecules escape into vapour phase, increasing vapour pressure.

23 (c) **Vapour pressure of pure benzene decreases by addition of naphthalene**

**Explanation:**

Colligative properties depend only on the number of solute particles. Naphthalene is a non-volatile solute, so it lowers the vapour pressure of benzene — a colligative effect.

24. (b) **Vapour pressure of atmospheric constituents**

**Explanation:**

Atmospheric pressure is produced by the gases present in the atmosphere. It is the sum of the partial pressures (vapour pressures) of all atmospheric constituents like nitrogen, oxygen, carbon dioxide, water vapour, etc.

25. (b) According to Raoult's Law

$$\frac{P^0 - P_s}{P^0} = x_B \quad (\text{Mole fraction of solute})$$

$$x_B = \frac{0.8 - 0.6}{0.8} = 0.25.$$

26. (d)  $\frac{P^0 - P_s}{P^0} = \text{molality} \times (1 - \alpha + x\alpha + y\alpha)$  the value of  $P^0 - P_s$  is maximum for  $BaCl_2$ .

27. (a)  $\frac{P^0 - P_s}{P^0} = \frac{18 \times 18}{180 \times 90} = 0.02.$

28 (c) **Raoult's law**

**Explanation:**

According to Raoult's law:

$$\frac{P^0 - P}{P^0} = X_B$$

The relative lowering of vapour pressure equals the mole fraction of the solute.



**29 (b) 0.365 methanol, 0.635 ethanol**

Explanation:

Total vapour pressure:

$$P_{\text{total}} = 2.619 + 4.556 = 7.175 \text{ kPa}$$

Mole fraction of methanol in vapour:

$$Y_{\text{methanol}} = \frac{2.619}{7.175} \approx 0.365$$

Mole fraction of ethanol in vapour:

$$Y_{\text{ethanol}} = \frac{4.556}{7.175} \approx 0.635$$

Therefore, vapour composition is:

- Methanol = 0.365
- Ethanol = 0.635

**30. (d)**  $P_T = P_P^0 X_P + P_Q^0 X_Q; P_T = 80 \times \frac{3}{5} + 60 \times \frac{2}{5}$

$$P_T = 48 + 24 = 72 \text{ torr.}$$

**31. (c)**  $\frac{P^0 - P_s}{P^0} = \frac{\frac{w}{m}}{\frac{w}{m} + \frac{W}{M}} \because \frac{W}{M} > \frac{w}{m} \Rightarrow \frac{640 - 600}{640}$

$$= \frac{w}{m} \times \frac{M}{W} \Rightarrow \frac{40}{640} = \frac{2.175 \times 78}{m \times 39.08}; m = \frac{2.175 \times 78}{39.08} \times \frac{640}{40}$$

$$m = 69.45.$$

**32 (a)**

Explanation:

Relative lowering of vapour pressure equals the mole fraction of solute. Therefore,

$$\frac{p - p_s}{p} = X_{\text{solute}} = \frac{n}{N + n}$$



33. (c) The lower is boiling point more is vapour pressure; boiling point order,  $HCl < HBr < HI < HF$

34. (a) **Mole fraction of solute**

Explanation:

According to Raoult's law, the relative lowering of vapour pressure is equal to the mole fraction of the solute.

$$\frac{P^0 - P}{P^0} = X_B$$

where:

- $P^0$  = vapour pressure of pure solvent
- $P$  = vapour pressure of solution
- $X_B$  = mole fraction of solute

35. (c)  $\frac{P^0 - P_s}{P^0} = \frac{n}{N} \Rightarrow \frac{P^0 - P_s}{P^0} = \frac{1}{9.9} \Rightarrow 9.9P^0 - 9.9P_s = P^0$

$$8.9P^0 = 9.9P_s \Rightarrow P_s = \frac{8.9}{9.9}P^0 \approx 0.90P^0$$

36. (a) **210.2**

According to Raoult's law:

$$\begin{aligned} \frac{P^0 - P}{P^0} &= \frac{n_B}{n_A} \\ \frac{0.30}{17.54} &= \frac{(20/M)}{100/18} \\ 0.0171 &= \frac{20 \times 18}{100M} \\ 0.0171 &= \frac{3.6}{M} \\ M &= \frac{3.6}{0.0171} \approx 210.5 \end{aligned}$$

37. (c) **45.24**

Relative lowering of vapour pressure:

$$\frac{195 - 192.5}{195} = \frac{2.5}{195} = 0.01282$$



Using:

$$\frac{P^0 - P}{P^0} = \frac{w_B M_A}{W_A M_B}$$

where:

- $w_B = 1 \text{ g}$
- $M_A = 58$
- $W_A = 100 \text{ g}$
- $M_B = ?$

$$0.01282 = \frac{1 \times 58}{100 \times M_B}$$

$$M_B = \frac{58}{1.282} \approx 45.24$$

38. (a) 1000 ml of  $\text{CH}_3\text{OH}$  requires methanol = 32 g.

150 ml of 2 M  $\text{CH}_3\text{OH}$  requires methanol

$$= \frac{32}{1000} \times 150 \times 2 = 9.6 \text{ g.}$$

39. (b)  $\therefore P^0 - P_s = P^0 \times \text{mole fraction solute}$

$$10 = P^0 \times 0.2; 20 = P^0 \times n \Rightarrow n = 0.4 \therefore N = 0.6.$$

40. (b) According to the Raoult's law for the non-volatile solute the relative lowering of vapour pressure of a solution containing a non-volatile is equal to the mole fraction of the solute.

41. (b) **A non-electrolyte in dilute solution**

Explanation:

Raoult's law is accurately applicable for dilute solutions of non-electrolytes because such solutes do not dissociate into ions.

42. (a) 0.25

Explanation:

According to Raoult's law:

$$X_B = \frac{P^0 - P}{P^0}$$



$$X_B = \frac{0.80 - 0.60}{0.80} = \frac{0.20}{0.80} = 0.25$$

43. (d) Relationship between mole fraction of a component in the vapour phase and total vapour pressure of an ideal solution.

$$y_A = \frac{P_A}{P_{total}} = \frac{x_A \cdot P_A^0}{x_A \cdot P_A^0 + x_B \cdot P_B^0}$$

$$= \frac{1 \times 1}{1 \times 1 + 2 \times 2} = \frac{1}{1+4} = \frac{1}{5} = 0.2$$

44. (c) Lowering in weight of solution  $\propto$  solution pressure

Lowering in weight of solvent  $\propto P^0 - P_s$

( $\because P^0$  = vapour pressure of pure solvent)

$$\frac{P^0 - P_s}{P_s} = \frac{\text{Lowering in weight of solvent}}{\text{Lowering in weight of solution}}$$

$$\frac{P^0 - P_s}{P_s} = \frac{w \times M}{m \times W}$$

$$\frac{0.05}{2.5} = \frac{10 \times 18}{90 \times m} \Rightarrow m = \frac{2 \times 2.5}{0.05} = \frac{2 \times 250}{5} = 100$$

